Team Delta: Capstone Project Proposal

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03/21/2026

Basics

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Preliminary Title:

Gunshot Residue Classification: Accuracy, Interpretability, and Failure Analysis

Background and Question

1) Background
   1.1) Domain Importance

   Gunshot residue analysis (GSR) plays a critical role in criminal investigations involving firearm discharge and firearms being used as main weapons in murder cases. When a gun is fired, microscopic particles are being released. The composition of these particles, such as lead, barium, and antimony, is a key indicator of gunshot residue and can function as major supporting evidence in court, helping investigators determine whether a suspect has discharged a firearm or has been in close proximity to one. Since GSR evidence can heavily influence legal outcomes, the reliability, validity, and interpretability of this data as evidence carries significant importance.

   1.2) The Problem

   Despite its importance, GSR analysis is often not straightforward. Traditional forensic methods rely on electron microscopy and spectroscopy such as Scanning Electron Microscopy (SEM) or Energy-Dispersive X-Ray Spectroscopy (EDS) to identify the individual particles based on their appearance and elemental composition. However, several other sources such as brake dust, industrial materials, fireworks, and other regular environmental exposures can show similar chemical signatures, increasing the likelihood of false positives where non GSR particles are identified falsely as GSR particles. Since interpretation relies on expert judgement, further variability is introduced into the final judgement. This raises concerns about the validity across analyses, which is a particular concern in high stakes fields like forensics.

1.3) Why ML Matters (Need/Niche)

Instead of relying on manual analysis and interpretation, machine learning models offer an alternative approach to GSR analysis that can improve consistency, scalability, and robustness of particle classification. Models can be specifically trained to minimize false positives and several methods exist that can provide interpretable results for model decisions, such as GradCAM plots for neural networks doing Computer Vision on images created from microscopy or SHAP analysis for tree based methods. Additionally, careful analysis can lead to a better understanding of where models struggle and fail, as well as features that contribute to misclassifications. This focus ensures that ML methods can be applied responsibly in forensic contexts and remain interpretable, which is paramount in fields like criminal justice (Rudin 2019).

1.4) Novelty

Applying ML to GSR analysis is not a novel concept in and of itself. Prior work emphasizes general model performance and efficiency (see Menking-Hogatt 2021; Mandel et al. 2018). Our focus, however, is novel in that we are planning to focus on understanding the underlying causes of model error, behavior, and interpretability. Rather than relying purely on model accuracy, we aim to identify which features contribute to classification decisions and analyze model failure modes. This approach represents a meaningful extension of existing work and is particularly valuable for high stakes decision making where false positives have disastrous consequences.

2) Question

2.1) Primary

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar particles?

2.2) Comparative

How does classification performance vary across different machine learning approaches when applied to gunshot residue identification?

2.3) Interpretability

Which elemental combinations contribute most strongly to model predictions, and how consistent are feature importance patterns across different models?

2.4) Failure Analysis

Under which conditions do machine learning models misclassify particles, particularly in cases where environmental particles are incorrectly labeled as gunshot residue? What does this reveal about the limitations of automated classification approaches?

3) Hypotheses and Predictions

3.1) Hypotheses

First, we hypothesize that machine learning models can effectively distinguish between true gunshot residue particles and environmental confounders. Model performance and reliability will depend on algorithm choice as well as the underlying chemical characteristics of particles.

Second, we hypothesize that non-linear methods will outperform linear methods since the relationships between elemental features in GSR are likely complex, involving interactions.

Third, we hypothesize that samples lacking strong indicators of GSR, such as lead, barium, and antimony, are more likely to be classified as false negatives, resulting in samples with atypical compositions to contribute to classification errors. Following, we also hypothesize that confounders such as brake dust contribute to higher rates of false positives due to greater overlap in elemental composition.

3.2) Predictions

Based on our hypotheses, we expect several outcomes from our analysis.

First, we expect non-linear models to outperform linear models in key metrics such as overall accuracy, recall, precision, and F1 score, with improvements driven largely by their innate ability to capture nonlinear relationships and interactions. However, we still expect significant hyperparameter tuning to reduce false positive rates, particularly in cases of chemically similar particles.

Second, we expect feature importance to show that lead, barium, and antimony presence is the most influential variable for classification. In addition, specific combinations or ratios of these elements are likely to emerge as key predictors of GSR, especially in borderline cases.

Finally, we expect misclassification to be driven largely by overlap in elemental composition of environmental confounders that do not fully match the classic Lead-

Barium-Antimony signature but are showing close composition to it. We expect these errors to highlight edge cases which would warrant further expert analysis.

Stakeholders

1) Our primary stakeholders are courts who require GSR analysis to support decision making. Results must be accurate, reliable, reproducible, and interpretable.
2) Our secondary stakeholders are expert witnesses and forensic analysts who may start with our pipeline and then dig deeper into certain aspects of the analysis. For those stakeholders, transparency and ease of use are paramount to support existing workflows and aid them in writing up expert testimony for court cases.
3) Defendants may also be stakeholders in that they are directly impacted by errors, making false positives a critical concern and raising concerns about transparent and fair ML. Defense counsel may hire their own expert witnesses or dig into failure modes and false positives to support their case.
4) From a funding standpoint, federal research organizations such as the National Institute of Justice (NIJ) would be an ideal stakeholder for financial support. As one of the largest financial backers for forensic science research, the NIJ could provide a grant for the study.

Data and Methods

1) Data
   1.1) Datasource #1
   Our primary datasource is the NIST gunshot residue (GSR) dataset (Ritchie & Reynolds 2021). This dataset contains particle-level measurements obtained via electron microscopy with energy dispersive X-ray spectroscopy (SEM/EDS). Each observation

represents a single particle and includes additional elemental composition data such as relative concentrations of barium, lead, and antimony along with the ground truth label (GSR/Confounder). The dataset is small, containing 30 discrete samples, but relatively balanced at 12 brake dust samples, 10 fireworks samples, and 8 GSR samples. It also contains metadata as well as TIFF images with embedded X-ray spectra for each particle. The samples come from the Boston Police Department and were taken from a volunteer who fired guns at a firing range. This data has several caveats. First, it only contains 30 discrete samples, which is not ideal for ML based approaches. Second, all samples come from a single volunteer at a firing range, reducing generalizability across modes and actors. Despite these issues, the NIST data provides a high-quality verified GSR dataset for benchmarking.

In addition to the NIST data, this project will explore integration of a publicly available GSR dataset maintained by the Netherlands Forensic Institute (NFI) (Matzen et al. 2022). Their GitHub repository contains SEM/EDS particle measurements of GSR and related particles, as well as classification labels designed for ML analysis. This data was collected from a variety of sampling locations between 2015 and 2019 and contains a total of 1953 samples from 210 criminal cases across 964k particles. Samples are organized into the same shot (n = 1.5k), or into different shots (n = 859k). Despite its larger size, the NFI data also has several caveats. Samples come from multiple case types and projects, which introduces batch effects and noise. It also includes only select elemental compositions, which makes it prone to missing edge case confounders or non-standard GRS particles. Despite its issues, the NFI data provides us with scale and diversity that can help harden our models and make them more robust.

1.1.1) Data Dictionaries: Target & Predictor Variables

Across both datasets, our target variable is a binary indicator of a GSR sample versus a non-GSR sample. In the NIST data, this is contained in the *gsr_label* field, whereas the NFI data uses the *merged_relevance_class* field.

Both datasets have a rich set of predictor variables, consisting primarily of different elemental compositions, indicating chemical signatures. The datasets also contain information about the ammunition type, case type (research or case), as well as TIFF images of the samples in the NIST case. We plan to use those as a secondary modality that may provide complementary information not fully captured in the tabular data.

1.1.2) NIST

1. Tabular

Column	Description
particle_id	Unique particle id
sample_id	Sample id (1-30)
sample_type	GSR, brake dust, firework
gsr_label (target)	Binary (1/0)
Pb, Ba, Sb	Respective elemental concentrations

2. Other

TIFF_image	Path to TIFF spectrum
metadata	Analysis configuration

1.1.3) NFI

The NFI data contains four tables. Below are the dictionaries per table.

1. Stub

Column	Description
id	Unique stub id
sample_name	Description of sample
project	Name of case or research project
type_project	Origin (case or research)
ammo_type	Ammunition type
sample_type	Type of sample
sample_source	Source of sample

2. Particle

Column	Description
stub_id	Stub id
particle_id	Unique particle id
relevance_class	Assigned particle class
merged_relevance_class (target)	Grouped particle class (particle-level GSR label)
Ac to Zr	Percentage of composition of elements in particle

3. Source

Column	Description
id	Unique source id

name	Name of source
cluster	Name of cluster
group_by	Method for grouping stubs

4. Stub Source

Column	Description
id	Unique id for stub-source combination
stub_id	Stub id
source_id	Source id from source table

2) Methods

2.1) Data Preprocessing

Prior to model development, several data preprocessing steps are necessary to ensure both datasets are compatible and ready for analysis.

We will first complete data cleaning and standardization where missing values will be assessed and either imputed where valid or dropped and units and scales of elemental concentrations will be standardized to match between NIST and NFI data. TIFF images will be processed to extract numerical features for additional computer vision approaches to modeling, such as CNNs.

We will also add net new features such as elemental ratios or morphology descriptors, if available.

Analysis Plan

1) Models

We are planning to analyze a comprehensive suite of different classification models spanning both linear and nonlinear methods, including deep learning. Our main linear baseline classifier will be a logistic regression to establish a performance floor. We will then build up in complexity using gradient boosted tree based approaches (XGB), as well as neural network (NN) architectures for tabular data and image data with iterations on hyperparameter tuning. Key parameters are regularization strength for logistic regression to give insight into initial feature selection. For gradient boosted methods, key parameters are tree depth, learning rate, number of estimators, and potential class weighting after analysis of class distribution to balance model complexity with number of false positives. Similarly, for neural networks we plan to tune depth, hidden layer size, learning rate, batch size, and regularization parameters (e.g. dropout, early stopping). All hyperparameter tuning will be done to pull models towards the desired calibration of low false positives balanced with low false negatives to ensure our models are both accurate and reliable in practice.

For tabular data, small fully connected NNs will be tested. For TIFF Images containing X-ray spectra, features will be extracted prior to modeling to be added to the existing tabular data. This will allow us to leverage spectral information in the TIFF images in addition to the existing tabular data.

While dimensionality reduction does not seem immediately necessary given the relatively small predictor size, the most likely use case for it are the elemental features (Ac-Zr) in the NFI data. We will explore whether dimensionality reduction techniques are beneficial for our task, starting with PCA for linear space and UMAP for non linear space. Because interpretability is central to our research question, however, dimensionality reduction will be applied selectively and compared to methods trained on the original feature space. Performance gains should not come at the expense of losing interpretability.

2) Evaluation

To assess model performance and answer our research questions, multiple evaluation metrics will be used.

Classification metrics will focus on accuracy, precision, recall, F1 score, and ROC-AUC. We will also use cross validation to avoid overfitting and provide robust estimates of model performance across folds. Comparative analysis between modeling approaches will help us identify the best performing method for GSR classification.

Central to our question is specificity, or false positive rate. Incorrectly labeling a sample as having GSR residue when it does not have any represents disastrous model failure. Because we expect imbalance in our classes, Precision-Recall AUC might better reflect true performance than ROC-AUC curves alone. Since we are interested in failure modes, confusion matrices will be heavily incorporated to dive deeper into misclassifications.

Finally, we aim to do cross-dataset validation where we train on the NFI data and test on the NIST data, ultimately aiming to show true model generalizability “in the wild” which will show how reliable our approach truly is.

3) Interpretability

Understanding why models make a decision is critical in forensic contexts. Feature importance can help us identify which elemental concentrations and ratios contribute significantly to classification. For tabular NNs, techniques such as SHAP or permutation importance can help quantify the influence of each feature on final model predictions. Additionally, we plan to look at integrated gradients to probe neuron activation based on input features. For the image derived features, contribution scores for spectral peaks can be analyzed. These techniques will help us answer which factors contribute most to misclassifications as well as correct classifications and turn “black box” models (or models that are traditionally thought of black box) into interpretable powerhouses for forensic contexts.

4) Misclassification Analysis

Investigating model failure is a critical point of our analysis. Our goal is to identify patterns in elemental composition or spectral features that lead to both false positives as well as false negatives, as both are costly in the courtroom. We will also pay particular attention to the effect of chemically similar particles and edge cases that warrant further expert analysis. These insights will inform both model limitations and forensic implications in practical GSR analysis.

5) What defines success (how do we know we answered the question)

Success will be measured across several criteria. First, models that achieve higher than baseline accuracy will be evaluated further for acceptable performance as determined by reliability, not just accuracy.

We also want to create models that generalize well across both NIST and NFI data, demonstrating that predictions are not dataset specific and analyze their key features and misclassification patterns.

By combining quantitative metrics, interpretability, and error analysis, this project aims to provide a rigorous assessment of the potential of ML in forensic GSR analysis.

Data Limitations

1) Sample Size and Independence

   The NIST dataset only has 30 samples, which is small for ML. Particles within each sample also aren't independent since they come from the same firing event or material, so splitting at the particle level during cross-validation would leak information and inflate performance. We'll use GroupKFold with sample/stub ID as the grouping variable to prevent this.

2) Dataset Integration and Batch Effects

   Combining the NIST and NFI datasets introduces batch effect risks since they were collected with different instruments, protocols, and contexts. The NFI data alone may already have this problem given its mix of case types. Models could learn to distinguish dataset origin instead of actual particle type, so we'll check whether dataset source shows up as an important feature and do cross-dataset validation.

3) Generalizability

   As previously mentioned, generalizability is also a concern. The NIST data comes from one volunteer at one firing range with specific ammo, and real-world GSR varies a lot by weapon, ammo brand, environment, and collection method. The NFI data adds breadth across 210 cases but only covers a limited set of elements.

Modeling Pitfalls

1) Overfitting

   Given the small size of the NIST dataset, complex models such as neural networks risk overfitting, where near-perfect training performance does not translate to held-out data. This risk will be monitored by tracking the gap between training and validation performance across all models. Starting with simpler models (logistic regression) before increasing complexity (XGBoost, neural networks) provides a natural check, as large jumps in training performance without corresponding validation improvement would signal overfitting.

2) Class Imbalance

   Although the NIST data is roughly balanced at the sample level (8 GSR, 12 brake dust, 10 fireworks), the particle-level class distribution may be skewed. The NFI dataset, with approximately 964,000 particles, may exhibit significant imbalance as well. If not addressed, models may default to predicting the majority class and achieve high overall accuracy while performing poorly on the minority class. Techniques such as class weighting, SMOTE, or threshold tuning will be explored to ensure balanced performance across classes.

3) Feature Engineering Risks

   Engineered features such as elemental ratios (e.g., Pb/Ba, Sb/Ba) introduce the possibility of extreme or undefined values when the denominator is zero or near-zero. These edge cases could produce outliers that disproportionately influence model behavior. All feature engineering decisions, including how zero-denominator cases are handled (e.g., small constant addition, imputation, or exclusion), will be documented and justified.

Interpretability Pitfalls

1) Feature Correlation and SHAP Instability

   Elemental concentrations in GSR particles tend to be correlated, which is a problem for SHAP. When features are correlated, SHAP can split importance between them inconsistently, so one model might point to barium while another points to lead when they're really capturing the same signal. Molnar (2022) notes that KernelSHAP assumes feature independence, producing unreliable results when features are dependent. Ordóñez-Muñoz et al. (2025) raise the same concern and recommend checking correlations before running SHAP. We'll assess correlations up front and may use grouped SHAP values or feature selection.

Forensic and Real-World Considerations

1) Base Rate and Predictive Value

   Reporting accuracy alone can be misleading in forensic contexts. If GSR particles are rare relative to environmental particles in a real sample, false positives can outnumber true positives even with a good model. This is the base rate fallacy (Thompson & Schumann, 1987). For example, 95% accuracy with a 5% false positive rate in a population where only 1% of particles are GSR would produce more false positives than true positives. Kaye (2016) argues that courts need to consider the base rate alongside the false positive rate when evaluating forensic evidence, not just accuracy.

2) Courtroom Implications

   Continuing that thought, ML model outputs must be communicated carefully in legal contexts. Overconfident predictions or poorly calibrated probability estimates could mislead investigators, attorneys, or juries. A specific example of Thompson & Schumann's base rate fallacy is a class of reasoning errors they termed the prosecutor's fallacy, where people confuse the probability of a match given innocence with the probability of innocence given a match. Saying a model is 95% accurate could easily be misread as a 95% chance that a flagged particle is truly GSR, when the actual positive predictive value could be much lower. We'll examine model calibration and present results with appropriate uncertainty.

Technical Details

1) Languages

   Python will be the primary language used for analysis and modeling due to its robust library support for various ML models and scientific computing, as well as the convenience of Jupyter Notebooks for scripting ML models and data analysis.
   SQL may be used (such as DuckDB) if larger datasets require complex joins.

2) Libraries

   Key libraries are: pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, pytorch, shap, opencv, and tifffile.

3) Data Visualization

   For complex analysis and ML support, Python libraries (matplotlib, seaborn, etc.) will be used. For general story telling in the final report, PowerBI or Tableau may be used.

4) Source File Types

   We expect raw datasets will primarily be CSV. However, if compression is needed for cleaning, prepping, and exploring excessively large datasets, conversion to Parquet may be done.

5) Version Control

   All code, preprocessing scripts, and model scripts will be version-controlled and maintained using git via GitHub. The repository will contain:

   - Preprocessing scripts for tabular and TIFF data
   - Feature engineering scripts
   - Model training and evaluation scripts
   - Visualization and interpretability analyses
   - Written documentation in markdown, readme, and pdf files

GitHub Repository

https://github.com/bkoconnell/datascience-capstone/

Citations

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Dataset Citations

NFI

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